VIKAS KOLHE

Bengaluru, India

+91 96375 20386 ⋄ vikas.kolhe.tech@gmail.com ⋄ linkedin.com/in/vikas-kolhe

PROFILE

Robotics Engineer and Embedded Systems Specialist with expertise in ROS2, AI vision systems, and IoT-based automation. Experienced in developing real-time robotics applications, optimizing hardware systems, and mentoring students in product development and innovation.

EXPERIENCE

Faculty - Technology Exploration and Product Engineering Feb 2026 - Present
GITAM University, Bangalore

- Teaching and mentoring students in product development from idea to prototype
- Working with Arduino, ESP8266, ESP32, and Raspberry Pi for real-world applications
- Guiding students in embedded systems, IoT, and robotics-based projects
- Supporting innovation through hands-on learning and rapid prototyping
- Utilizing programming skills in Python and Embedded C++ for development and teaching

Senior Robotics Associate Aug 2024 - Feb 2026
OMOTEC (On My Own Technology), Pune

- Developed robotics applications using ROS2 Humble on Raspberry Pi and ESP32.
- Built AI vision systems using OpenCV and TensorFlow Lite with 94% accuracy at 30 FPS
- Designed motor control systems with PWM and PID tuning improving precision by 28%
- Implemented SPI, I2C, UART communication achieving sub-10ms latency
- Optimized PCB designs reducing cost and improving signal integrity by 40%
- Delivered STEM kits to 15+ schools generating significant revenue and engagement

Robotics Trainer Jul 2023 - Jul 2024
STEMROBO Technologies, Mumbai

- Trained 500+ students across 25 institutions with 92% satisfaction rate
- Developed FreeRTOS training modules with 78% knowledge retention
- Conducted 18+ workshops on IoT, Embedded Systems, and Robotics
- Mentored teams achieving top ranks in national competitions
- Built partnerships with schools impacting 2000+ students annually

PROJECTS

Smart Pothole Detection and Repair System 2023
Raspberry Pi 5, Python, OpenCV

- Developed real-time pothole detection system using computer vision
- Automated repair mechanism for efficient road maintenance

ESP32-Based Humanoid Robot 2023
ESP32, Embedded C++, IoT

- Designed humanoid robot with Wi-Fi control and servo coordination
- Integrated OLED display for interactive expressions

EDUCATION

B.Tech in Electronics & Telecommunication 2023
Dr. Babasaheb Ambedkar Technological University, Pune
CGPA: 8.35/10

Diploma in Electronics & Telecommunication 2020
MSBTE, Mumbai
87.41%

SKILLS

Programming	Embedded C, C++, Python
Platforms	ROS2, Arduino, ESP8266, ESP32, Raspberry Pi
Tools	OpenCV, TensorFlow Lite, MATLAB
Protocols	UART, SPI, I2C
Concepts	Embedded Systems, IoT, Robotics, AI Vision

DECLARATION

I hereby declare that the above information is true to the best of my knowledge.